Nonlinear wave physics

Adversarial novelty review | Evidence cutoff: 14 October 2026 | Paid API spend: $0

Decision: HOLD on committing 8-12 weeks. There is a credible, narrow mechanism question, but no verified original discovery yet. Permit one bounded validation gate. Stop treating recurrence, torus formation, hysteresis or an 8.34% truncation shift as sufficient novelty.

The strongest immediate question is whether the modulation below the locked-orbit instability is a separate attracting invariant set, and how its branch connects to the periodic orbit. A useful contribution would explain that connection and its modal cause across parameters. Finding another example of a familiar bifurcation would be a limited application.

What was independently checked

I read the supplied findings, handoff, governing equations, scripts and diagnostics. The model is the fixed-end, uniformly forced alpha-FPUT chain:

q_ddot[n] = r[n] - r[n-1] + alpha (r[n]^2 - r[n-1]^2)
- eta q_dot[n] + F cos(Omega t),
r[n] = q[n+1] - q[n]; q[0] = q[N+1] = 0.

Here N=8, alpha=0.25, eta=0.0025 and Omega=0.347. Eight modes are an exact coordinate transformation; four modes are a different, projected dynamical system. F is the same per-site force amplitude in both.

Independent audit	Full eight-mode chain	First four modes
Critical force F (rounded)	0.0046931582	0.0050846446
Leading multiplier at crossing	0.984914 + 0.173044 i	0.983327 + 0.181847 i
d|mu|/dF (finite difference)	17.92	21.19
Largest |mu| at F=0.0045	0.9961540	0.9776203
Largest |mu| at F=0.005	1.0048306	0.9981292

The audit used separately coded bond forces and Hessians, physical coordinates for the full chain, DOP853 (rtol=2e-11, atol=2e-13), and shooting. Residuals were below 2.4e-14. The monodromy identity M^T J M = exp(-eta T) J held within 4.2e-12. Both crossings agree with the package within 1e-9 in F; these are numerical checks, not rigorous enclosures. All four supplied tests also passed. Audit runtime was about 26 seconds.

Important reinterpretation: the package already records four-mode modulation at F=0.005, where its locked orbit is stable. Thus neither Floquet crossing is automatically the onset of accessible modulation. The four-mode model may have its own coexistence region. I did not independently rerun the long trajectories.

What the older literature already covers

Primary-source comparison. Numbers refer to linked references on page 5.

Paper and evidence	Actual result and comparison with this package
Shi & Ren (2026) [1] **Full March 2026 manuscript** + publisher record	Same model and representative parameters. Reports low-mode energy modulation, jumps and recurrence windows, using zero-state starts. It does not supply periodic-orbit Floquet continuation or classify coexisting invariant sets. Its discussion explicitly leaves analytical explanation challenging. The baseline modulation is replication; a branch-connection mechanism remains a candidate extension.
Geist & Lauterborn (1988), Part I [2] **Publisher metadata/abstract**	Damped, periodically driven Toda chain with periodic boundaries; periodic, quasiperiodic and aperiodic responses. This defeats a broad claim of newly discovering sustained nonperiodic response in a driven nonlinear chain. It does not establish our exact bifurcation.
Geist & Lauterborn (1990), Part II [3]; Geist, Parlitz & Lauterborn (1990) [4] **Part II abstract; methods full text**	Part II analyzes tori with Fourier and Lyapunov tools. The related primary methods paper, Sec. 5, specifies N=15 periodic Toda, localized sinusoidal forcing and relative-velocity damping, with two-/three-frequency tori and strange attractors. These differ from our fixed ends and uniform drag/forcing. Lyapunov convergence is already a central concern.
Khomeriki, Lepri & Ruffo (2001) [5] **Full manuscript**	Periodic beta-FPU chain, staggered forcing near the upper band edge. Rotating-wave modal theory predicts locked-mode instability and compares with simulations. Near-zero starts can leave the stable branch's basin before its local instability. Its modulated standing waves are spatially modulated and time-locked: they are not evidence for our temporal torus.
Khomeriki, Lepri & Ruffo (2004) [6] **Full manuscript**	Boundary-driven beta-FPU, damping near the far boundary: conducting/insulating coexistence and distinct increasing/decreasing thresholds. FPUT hysteresis is established computationally. This transport mechanism does not settle locked-versus-modulated coexistence here. The bistability paper is 2004, not 2001.
Charalampidis et al. (2015) [7] **Full primary article**	Boundary-driven, damped Hertzian trimer chain. Newton shooting and monodromy locate saddle nodes and Neimark-Sacker instabilities; continuation tracks branches. Sec. 4.3 warns that stable multipliers close to unity produce misleading long transients. This is direct methodological precedent, with different springs and geometry.

These precedents explain why the observations are plausible under established nonlinear dynamics. They do not quantitatively explain the particular thresholds or prove the topology of this chain's attractors. General theoretical compatibility and an actual model-specific explanation are different standards.

Reductions and the Toda connection

A new model must explain something beyond an ordinary truncation or fitted threshold.

Paper and evidence	Actual result and relevance
Hofstrand (2024) [8] **Full accepted manuscript**	Conservative periodic alpha+beta FPUT tangent to Toda (beta=2 alpha^2/3). Long-wave calculations locate departure from Toda spectral invariance near a predicted shock time. Different dynamics, boundaries and asymptotic regime. Its Jacobi/Lax spectral Floquet analysis is not temporal monodromy stability of our forced orbit.
Bilman & Nenciu (2016) [9] **Full manuscript**	Hamiltonian perturbations of the doubly infinite Toda chain. Derives scattering-data evolution; computes perturbed solitons and emergence of discrete eigenvalues. This is substantial near-integrable prior art, not a theory of our dissipative attractors. The package's exact one-soliton benchmark is validation infrastructure.
Deng et al. (2026) [10] **Accepted-paper abstract only**	APS confirms acceptance on 2 October 2026. Diatomic cubic-potential FPUT: the authors report optical-acoustical-acoustical resonant triads and a reduced Fourier model reproducing recurrence. This two-branch mechanism is not directly available in our monatomic chain. Full text was inaccessible, so coefficients, assumptions and proofs were not independently audited.
Li, Jain & Haller (2022), Part I [11]; Li & Haller, Part II [12] **Full primary texts**	Spectral-submanifold reductions for forced mechanical systems with internal resonance; reduced equilibria/limit cycles predict periodic/quasiperiodic responses and bifurcations. Generic reduction plus torus prediction is already developed. Theorems require assumptions and distinguish truncated dynamics from omitted-term persistence; they do not automatically certify this finite-amplitude FPUT case.

What is established, and what is not

Established mathematics: a transverse complex Floquet crossing is a candidate Neimark-Sacker bifurcation. Nonresonance and a nonzero first Lyapunov coefficient are needed for the generic local classification. The coefficient and crossing orientation determine criticality; coexistence alone does not. A local normal form cannot by itself locate a remote attractor or its basin. [12]

Existing computations: driven-chain multistability, tori, modal-transfer reductions and truncation errors have precedents. **Our addition so far:** reliable model-specific local thresholds and preliminary below-threshold modulation. **A new contribution would need:** a verified branch mechanism with explanatory parameter dependence, or a derived modal correction that makes independently successful predictions.

For the full crossing, arg(mu) predicts a local modulation scale of approximately 36.13 drive periods (four-mode: 34.36). This is an audit-derived onset prediction, not the modulation period far along a branch. Modal-energy harmonics can also differ from the fundamental state frequency.

Do not prioritize Toda scattering now. Conservative integrability neither survives ordinary forcing/damping nor follows from oscillatory modal energies. A future Toda link would need a compatible boundary/reference system and separation of forcing/damping-induced spectral drift from potential-induced drift.

Two questions worth testing

Conditional research opportunities, not certified open problems.

1. What invariant-set connection creates modulation below the locked-branch threshold? At F=0.0045, is there an attracting invariant circle of the stroboscopic map, and does its continuation connect to the F=0.004693158 crossing through a subcritical Neimark-Sacker bifurcation and a fold of circles? Other allowed outcomes are a disconnected branch, a resonant periodic orbit, chaos or a long transient.

Evidence distinction: coexistence and these bifurcation mechanisms are established possibilities [4-7,12]. Our evidence is one locally stable orbit plus finite-time modulation. A useful new result would establish the actual branch connection and explain how damping/detuning and modal couplings control it. Merely labelling a torus is insufficient.

Falsifiers: relaxation to the locked orbit defeats the claimed second attractor for that initial condition. A converged resonant cycle or positive Lyapunov exponent defeats the invariant-circle hypothesis. A supercritical coefficient defeats the proposed local subcritical scenario, although it does not rule out remote coexistence. A fold cannot be inferred from an upward/downward sweep alone.

2. Which omitted modal feedback causes the 8.34% local threshold shift? Can a coefficient-derived correction predict its size and criticality across independently chosen damping and detuning values?

Evidence distinction: modal reductions and normal-form corrections are mature [5,10-12]; this package has one discrepancy, not a new reduction principle. Seek a causal coupling pathway and predictive correction, not another fitted four-mode model. Uniform forcing directly projects onto odd modes, so deleting modes 5-8 also removes direct forcing of modes 5 and 7. All linear modes have the same decay rate eta/2; rapid slaving cannot be assumed from higher frequency alone.

Falsifiers: test symmetry-consistent groups of coupling coefficients, successive truncations and critical-mode participation. Derive the correction before testing held-out parameters. If selected couplings do not control the shift, or the correction fails outside the original point, reject the mechanism. Routine higher-order reduction with no further insight is an application result.

Next experimental decision

Recommend question 1 first; cap the gate at two working days / eight CPU-hours. At F=0.0045, use the zero state and perturbed locked states with independent physical-coordinate adaptive and refined fixed-step integration. Compare 3,500, 10,000 and 30,000-period diagnostics; save trajectories and actual drive phase. Check nearby fixed F values before claiming an interval of coexistence.

Classify the stroboscopic state using return structure, rotation/frequency convergence and a converged Lyapunov spectrum. An attracting invariant circle has one neutral map exponent and negative transverse exponents; a finite-time near-zero estimate alone is inconclusive. Compute the first Lyapunov coefficient and track the critical pair. Only then attempt invariant-circle continuation and a fold/connection test.

Advance only if coexistence survives numerical refinement and a concrete branch mechanism or predictive modal discrepancy remains to explain. A routine known scenario alone does not earn the full programme. Stop this novelty claim if it reduces to transient decay, numerical error, or standard replication; keep the verified solver.

Limitations: no full long-time rerun, basin map, torus continuation or Lyapunov-coefficient calculation was performed in this review. Geist originals and Deng full text remain access gaps. Searches and citation expansion through the cutoff cannot certify worldwide absence of prior work. The answer is a defensible candidate question, not proof that it is genuinely unresolved.

References and evidence trail

Primary papers; access status is recorded in the comparison tables.

[1] Y. Shi and H. Ren (2026). Recurrence in a periodically driven and weakly damped Fermi-Pasta-Ulam-Tsingou chain. Physical Review E 113, 034221. Published 26 March 2026.
Primary source / DOI | Open manuscript

[2] K. Geist and W. Lauterborn (1988). The nonlinear dynamics of the damped and driven Toda chain: I. Energy bifurcation diagrams. Physica D 31, 103-116.
Primary source / DOI

[3] K. Geist and W. Lauterborn (1990). The nonlinear dynamics of the damped and driven Toda chain: II. Fourier and Lyapunov analysis of tori. Physica D 41, 1-25.
Primary source / DOI

[4] K. Geist, U. Parlitz and W. Lauterborn (1990). Comparison of Different Methods for Computing Lyapunov Exponents. Progress of Theoretical Physics 83, 875-893. Sec. 5 verifies the Toda model and tori.
Primary source / DOI

[5] R. Khomeriki, S. Lepri and S. Ruffo (2001). Pattern formation and localization in the forced-damped Fermi-Pasta-Ulam lattice. Physical Review E 64, 056606.
Primary source / DOI | Open manuscript

[6] R. Khomeriki, S. Lepri and S. Ruffo (2004). Nonlinear supratransmission and bistability in the Fermi-Pasta-Ulam model. Physical Review E 70, 066626.
Primary source / DOI | Open manuscript

[7] E. G. Charalampidis et al. (2015). Time-Periodic Solutions of Driven-Damped Trimer Granular Crystals. Mathematical Problems in Engineering, article 830978.
Primary source / DOI

[8] A. Hofstrand (2024). Near-integrable dynamics of the Fermi-Pasta-Ulam-Tsingou problem. Physical Review E 109, 034204.
Primary source / DOI | Open manuscript

[9] D. Bilman and I. Nenciu (2016). On the evolution of scattering data under perturbations of the Toda lattice. Physica D 330, 1-16. Bilman's first name is Deniz, not G.
Primary source / DOI | Open manuscript

[10] G. Deng, A. Pezzi, G. Lin and M. Onorato (2026). Recursive behavior in a diatomic Fermi-Pasta-Ulam-Tsingou lattice. Physical Review E, accepted 2 October 2026; no final volume/page inferred.
Primary source / DOI

[11] M. Li, S. Jain and G. Haller (2022). Nonlinear analysis of forced mechanical systems with internal resonance using spectral submanifolds, Part I: Periodic response and forced response curve. Nonlinear Dynamics 110, 1005-1043.
Primary source / DOI

[12] M. Li and G. Haller (2022). Same series, Part II: Bifurcation and quasi-periodic response. Nonlinear Dynamics 110, 1045-1080.
Primary source / DOI

Package evidence: STAGE2_FINDINGS.md; NEXT_HANDOFF.md; stage1.py; floquet.py; refine.py; basin_check.py; floquet_refined.json; floquet_detailed.json; modal_summary.csv; test_stage2.py. The original package was left unchanged.

Audit reproducibility: form the 9-by-8 fixed-end bond matrix D, with r=Dq. Use force -D^T(r+alpha r^2)-eta p+F 1 cos(Omega t) and stiffness Jacobian -D^T diag(1+2 alpha r) D. Project with the first four sine modes for m=4. Solve P_F(y)-y=0, integrate its variational equation and find max|eig(M)|=1. Root brackets: [0.00468,0.00471] and [0.00507,0.00510]; crossing slope used F +/- 1e-6. Seeds came from floquet_detailed.json.